

Automated Deployment of a Static Website to AWS EC2 using GitHub Actions (CI/CD)

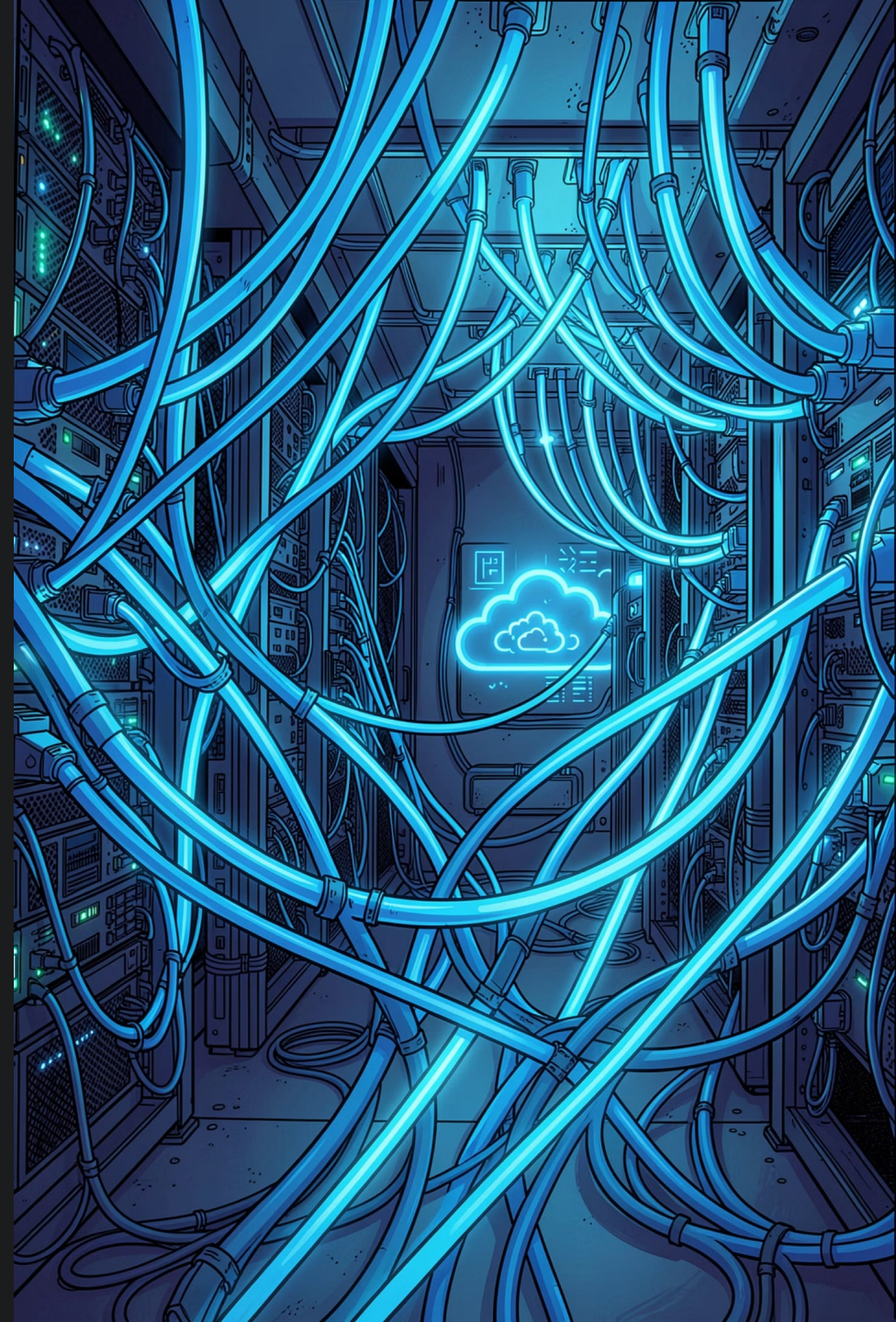
GIT

GITHUB ACTIONS

AWS EC2

APACHE

UBUNTU LINUX



Project Overview



Objective

Automate the deployment of a static website to an AWS EC2 instance using GitHub Actions — eliminating every manual step in the release process.

The Problem

- Manual server login and file transfer
- Manual web server restarts
- High risk of human error

The Solution

A fully automated CI/CD pipeline that deploys website updates **instantly** whenever code is pushed to GitHub.

Project Objectives

Automate Deployment

Build a CI/CD pipeline that deploys on every push — zero manual steps required.

Speed & Reliability

Ensure faster, consistent deployments with significantly reduced error rates.

Secure Integration

Connect GitHub to AWS EC2 securely using SSH key authentication.

Cloud & DevOps Skills

Gain hands-on experience with AWS, Apache, and industry-standard DevOps workflows.



Technologies & CI/CD Pipeline Workflow

Technology Stack

Git	Version Control
GitHub	Code Repository
GitHub Actions	CI/CD Automation
AWS EC2	Cloud Virtual Server
Ubuntu Linux	Operating System
Apache	Web Server
SSH / SCP	Secure Connection & Transfer

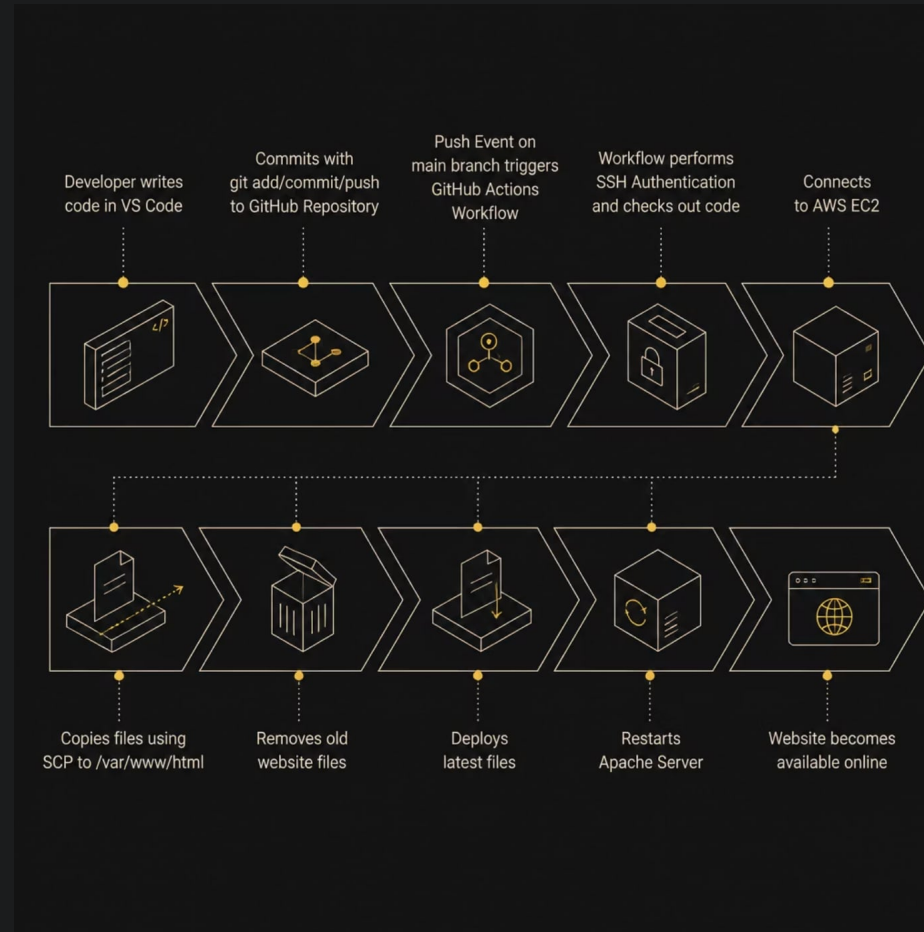
How the Pipeline Works

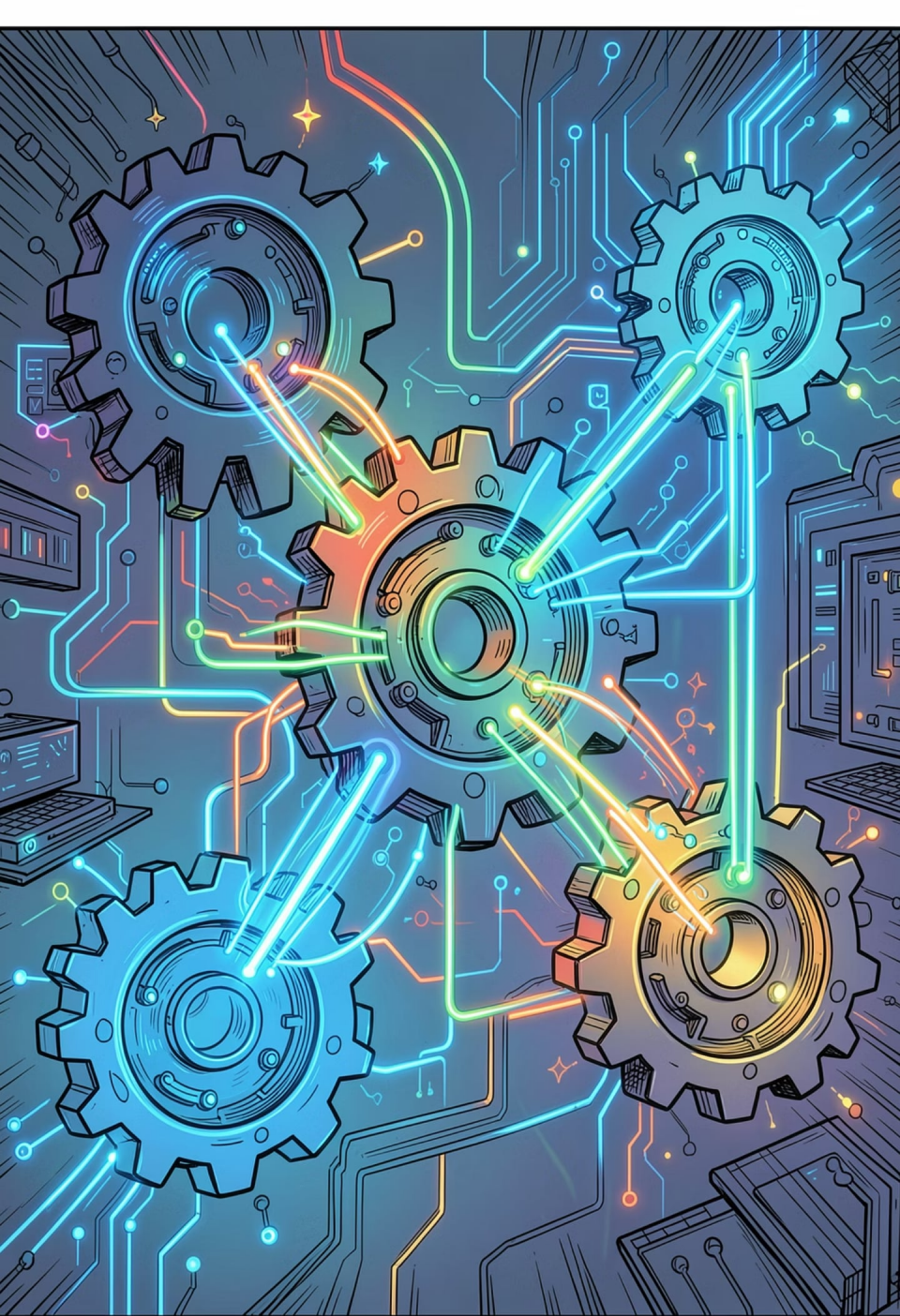
01	02
Developer writes and commits code	Code pushed to GitHub
03	04
GitHub Actions detects the push	Workflow starts automatically
05	06
GitHub connects to EC2 via SSH	Files copied using SCP
07	08
Old files replaced, Apache restarts	Updated website goes live

System Architecture

The end-to-end flow starts with the developer writing code in VS Code, committing changes via Git, and pushing to the GitHub repository. A push event on the **main branch** triggers GitHub Actions, which authenticates to AWS EC2 via SSH, transfers files using SCP to `/var/www/html`, replaces old files, and restarts Apache — making the updated website immediately available online.

Every step from code commit to live deployment is fully automated — no manual intervention required.





Why GitHub Actions?

GitHub Actions automatically triggers deployment workflows the moment code is pushed to your repository.



Faster Deployment

Workflows trigger instantly on every push, eliminating wait time.



Fully Automated

From checkout to Apache restart — the entire process runs without human input.



Reduced Errors

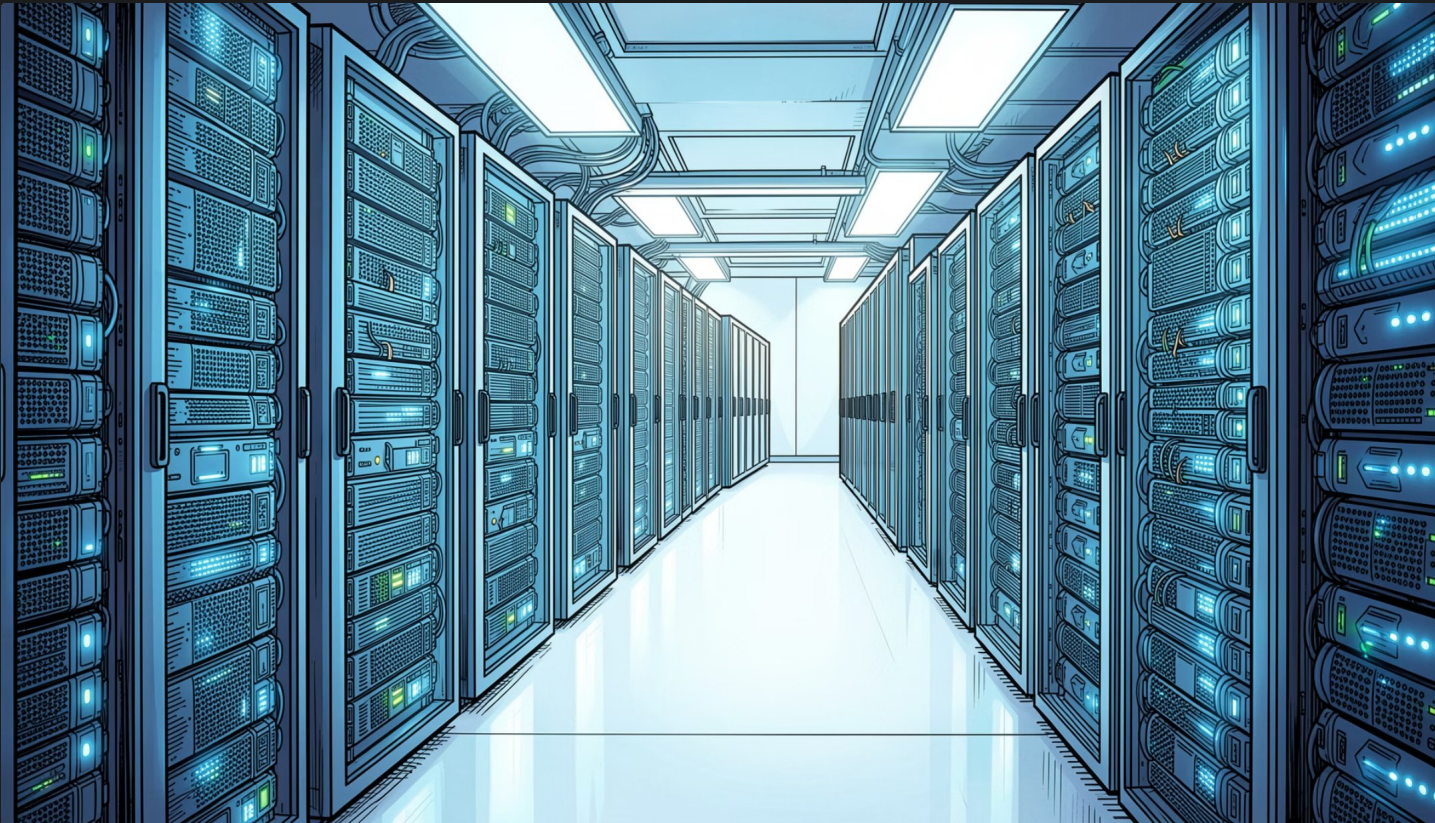
Consistent, repeatable deployments eliminate manual mistakes.



Native Integration

Works seamlessly within GitHub — no third-party CI/CD setup required.

AWS Deployment Infrastructure



AWS EC2

A virtual cloud server running **Ubuntu Linux** that hosts the static website. Provides scalable, on-demand compute capacity in the cloud.

Apache Web Server

Serves website files to visitors. Reads content from `/var/www/html` — the standard document root for Apache on Ubuntu.

Security Layers

- **SSH Key Authentication** — passwordless, secure server access
- **Security Groups** — firewall rules controlling inbound/outbound traffic
- **IAM Permissions** — least-privilege access control for AWS resources

Advantages of This Approach

Zero Manual Uploads

Every deployment is triggered automatically — no FTP clients, no SSH file transfers by hand.

Faster Software Delivery

Code reaches production in seconds after a push, dramatically shortening the release cycle.

Secure by Design

SSH key authentication and IAM roles ensure only authorized actions reach the server.

Cost-Effective & Scalable

AWS EC2 offers flexible pricing and easy scaling. GitHub Actions is free for public repositories.

Industry-Standard DevOps

This pipeline mirrors real-world workflows used by engineering teams at scale.

Results & Future Enhancements

✓ Project Outcomes

- Static website successfully hosted on AWS EC2
- Apache web server configured and serving content
- Automated deployment via GitHub Actions — Continuous Deployment achieved
- Version-controlled development workflow using Git

🚀 Future Scope

Docker &
Kubernetes

HTTPS with SSL

AWS Load Balancer

Auto Scaling

CloudWatch
Monitoring

Multi-Environment
(Dev / Test / Prod)

Conclusion

This project demonstrates a complete **DevOps deployment pipeline** by integrating Git, GitHub, GitHub Actions, AWS EC2, and Apache. Whenever new code is pushed, deployment executes **automatically** — ensuring faster, secure, and reliable website delivery with zero manual intervention.

Git Version Control	GitHub Code Repository	GitHub Actions CI/CD Automation
AWS EC2 Cloud Hosting	Apache Website Hosting	

📄 **Thank You** — Questions are welcome!